
EFFECTS OF UNILATERAL AND BILATERAL PLYOMETRIC TRAINING ON POWER AND JUMPING ABILITY IN WOMEN

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ABSTRACT

Makaruk, H, Winchester, JB, Sadowski, J, Czaplicki, A, and Sacewicz, T. Effects of unilateral and bilateral plyometric training on power and jumping ability in women. *J Strength Cond Res* 25(12): 3311–3318, 2011—The purpose of this study was to examine the effects of unilateral and bilateral plyometric exercise on peak power and jumping performance during different stages of a 12-week training and detraining in women. Forty-nine untrained but physically active female college students were randomly assigned to 1 of 3 groups: unilateral plyometric group ($n = 16$), bilateral plyometric group (BLE; $n = 18$), and a control group ($n = 15$). Peak power and jumping ability were assessed by means of the alternate leg tests (10-second Wingate test and 5 alternate leg bounds), bilateral leg test (countermovement jump [CMJ]) and unilateral leg test (unilateral CMJ). Performance indicators were measured pretraining, midtraining, posttraining, and detraining. Differences between dependent variables were assessed with a 3×4 (group $\times$ time) repeated analysis of variance with Tukey's post hoc test applied where appropriate. Effect size was calculated to determine the magnitude of significant differences between the researched parameters. Only the unilateral plyometric training produced significant ($p < 0.05$) improvement in all tests from pretraining to midtraining, but there was no significant ($p < 0.05$) increase in performance indicators from midtraining to posttraining. The BLE group significantly ($p < 0.05$) improved in all tests from pretraining to posttraining and did not significantly ($p > 0.05$) decrease power and jumping ability in all tests during detraining. These results suggest that unilateral plyometric exercises produce

power and jumping performance during a shorter period when compared to bilateral plyometric exercises but achieved performance gains last longer after bilateral plyometric training. Practitioners should consider the inclusion of both unilateral and bilateral modes of plyometric exercise to elicit rapid improvements and guard against detraining.

KEY WORDS single-leg exercises, double-leg exercises, depth jump, drop jump, performance enhancement, detraining

INTRODUCTION

Plyometric exercises are widely believed to lead to positive adaptations in terms of power production and corresponding improvements in tasks, which have shown a strong relationship with athletic performance such as sprinting and jumping. One possible mechanism explaining the effectiveness of plyometric exercises can be related to specific muscle performance in the stretch-shortening cycle (SSC) (3,12,28). One of the basic tenants of the SSC is that an activated muscle produces higher force in the concentric phase when immediately preceded by a rapid eccentric muscle action. Hence, such exercises, implemented particularly in jumps, are widely applied in contemporary training (20).

Jumping exercises can be grouped according to a number of criteria, one of these being whether they are performed in bilateral or unilateral fashion. Single-leg exercises can be performed either alternately, that is, using right and left legs in turns, or unilaterally by repeating with 1 leg only. It is common to see both methods used in varying amounts when training in many sports. As a rule, they are usually combined with strength exercises, likely because complex training has been demonstrated to lead to greater improvements in sport performance when compared to strength training or plyometric training alone (7,15). The fact that plyometric training is usually not performed in isolation might help to explain why previous attempts at examining both single and double-leg performance parameters have rarely been

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undertaken. The comparison of the effects of unilateral and bilateral plyometric training seems to be the more useful, because the differences between single plyometric exercises (e.g., 1-leg and 2-leg jumps) have been thoroughly identified and described in the literature (11,24,26).

Stannard (23) has indicated that single-leg plyometric training led to greater improvement in unilateral vs. bilateral jumping tests, whereas bilateral training led to a similar improvement occurring in both unilateral and bilateral tasks. Interestingly, combining single- and double-leg plyometric training did not lead to improvement in performance during a standing triple jump, whereas performing single- or double-leg jumps training in isolation did. Similar findings were reported by Delcore et al. (4), who also suggested that unilateral plyometric exercises are more effective in improving power abilities observed in bilateral tests. McCurdy et al. (18) combined unilateral and bilateral strength training with unilateral and bilateral plyometric exercises. They reported that in a unilateral test the unilateral exercise group improved more than the bilateral exercise group on relative power and vertical jump height. In a bilateral test, the improvements in power and jumping ability were similar in both groups.

The available body of literature suggests that a background knowledge of single exercise characteristics is helpful when planning for training programs. Therefore, the purpose of this study was to identify and compare the effects of unilateral and bilateral plyometric exercises on peak power and jumping ability demonstrated alternately, unilaterally, and bilaterally during 12 weeks of training followed by 4 weeks of detraining.

METHODS

Experimental Approach to the Problem

The study was designed to examine differences in peak power and jumping ability after unilateral (ULE) and bilateral (BLE) plyometric training and detraining. Subjects consisted of untrained female college students randomly assigned to 2 experimental training groups and 1 control group. The subjects in the ULE group underwent only unilateral plyometric exercises, whereas the BLE group performed bilateral exercises only. The plyometric programs involved analogous types of exercises (e.g., single vs. double-leg bounds) with the same number of contacts with the ground and with intensity controlled between groups. The dependent variables, peak power and jumping height and jumping distance, were measured 4 times: 3 days before the program (pretraining), during week 7 (midtraining), and 3 days (posttraining) and 4 weeks after the program (detraining). To verify the reliability of measures for chosen tests and familiarize the subjects with plyometric exercises the training program was preceded by pilot studies (3 weeks before testing) and familiarization training (6 sessions), respectively.

Subjects

Fifty-four untrained female college students volunteered to participate in this study. The subjects were physically active

TABLE 1. Characteristics of the training and control groups at pretraining (mean $\pm$ SD).*

	ULE	BLE	CON
Age (y)	20.6 $\pm$ 1.3	20.9 $\pm$ 1.7	21.0 $\pm$ 1.9
Height (m)	1.67 $\pm$ 0.4	1.66 $\pm$ 0.5	1.66 $\pm$ 0.6
Body mass (kg)	59.2 $\pm$ 4.9	57.3 $\pm$ 4.2	60.2 $\pm$ 5.2
Sum of skinfolds (mm)	56.4 $\pm$ 6.4	55.8 $\pm$ 6.9	57.7 $\pm$ 7.3

*ULE = unilateral plyometric group; BLE = bilateral plyometric group; CON = control group.

for 8–10 h·wk⁻¹ because of the nature of their studies (gymnastic, Nordic walking, swimming, and soccer). They were not involved in a strength training or conditioning program during this study, but they were experienced in explosive activities (i.e., basketball, volleyball, taekwon-do, and track and field). After meeting an attendance criterion (3 absences disqualified a participant from the study), 49 of them completed the study. All subjects were asked to abstain from any additional sport and recreation activities and to keep their diet unchanged for at least 6 weeks before and during the study. None of the subjects were taking any medications and nutritional supplements. Subjects signed an informed consent form before participating, and approval from the university's Ethics Committee was obtained before starting the study. Subjects were randomly allocated to 1 of 3 groups: unilateral plyometric (ULE; $n = 16$), bilateral plyometric (BLE; $n = 18$), and a control group (CON; $n = 15$). Descriptive characteristics for each group are shown in Table 1. The sum of skinfold thickness consisting of triceps, subscapular and suprailiac skinfold thickness, was measured using the methodology described by Heath and Carter (8). The values of researched parameters of the groups did not differ significantly ($p > 0.05$) before the program.

Testing

Peak power and jumping ability in alternate leg exercises were estimated by the Wingate test (10 seconds) and 5 alternate leg bounds, respectively. In bilateral exercises, they were assessed with the use of a countermovement jump (CMJ), whereas in unilateral exercises, a unilateral CMJ (UCMJ) was performed. The measurements were made 4 times: 3 days before the program, during the seventh week, and 3 days and 4 weeks after completion of the program. The Wingate test involved 2 trials with 8- to 9-minute rest in between, whereas 3 trials of jumping tests were separated by 1–2 minutes of rest. The best results (peak power and the furthest distance or the maximum height) from all the trials were used for the analysis. There was a rest period of at least 15 minutes between successive tests.

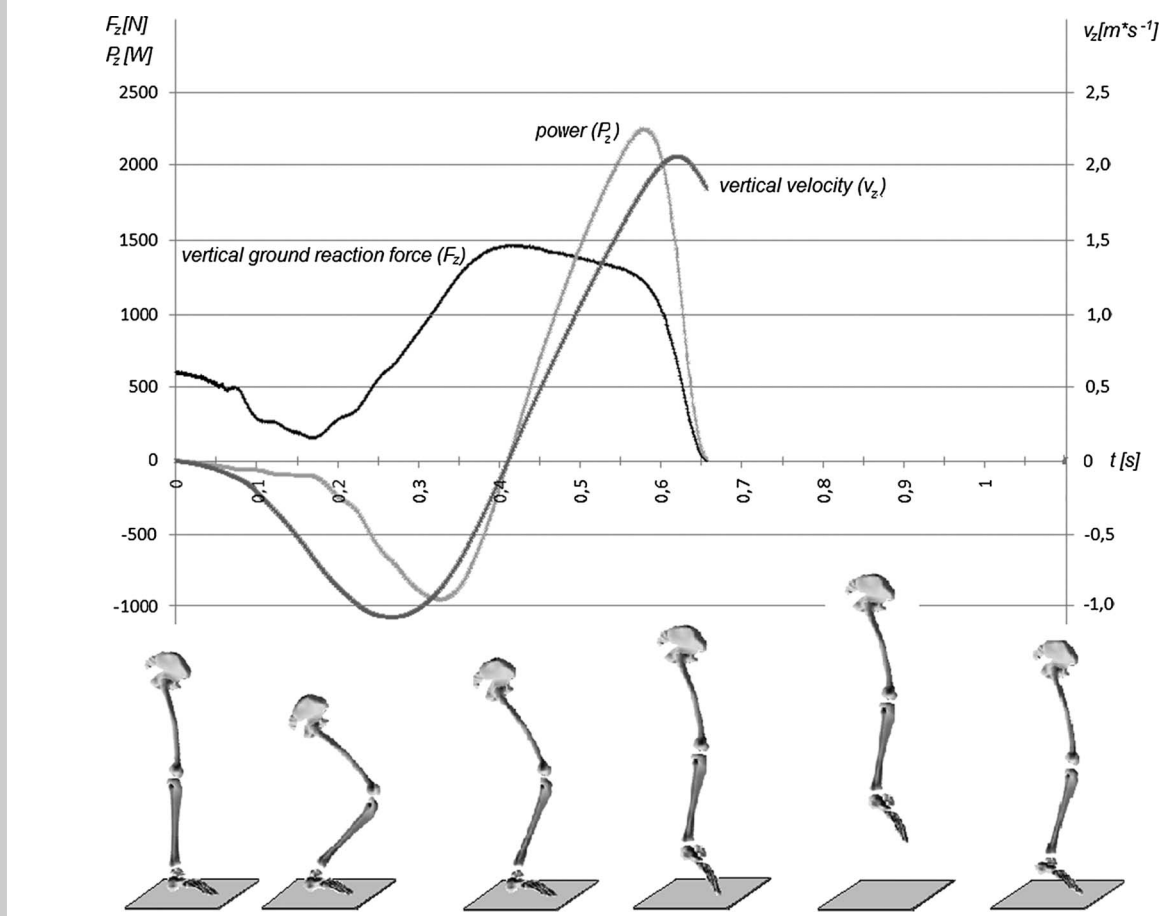


Figure 1. Countermovement jump—typical time courses of measured and computed parameters.

The Wingate test was performed in the morning, whereas the other tests were carried out in the afternoon. A 3-week test-retest study conducted before the training programs confirmed the reliability of the selected tests. The intraclass correlation coefficients were high, that is, ICCs = 0.90–0.95 for peak power and ICCs = 0.86–0.93 for jumping ability.

The Wingate test (10 seconds) was performed on a Monark cycle ergometer (834E, Varberg, Sweden) connected to IBM PC Pentium computer running the program MCE v. 4.5 (JBA Zb. Staniak, Poland) (22). Optical sensors were installed on a flywheel. A single pedal revolution was equal to 3.7 revolutions of the flywheel or 6 m. The test consisted of performing an exercise of a 10-second maximal effort on a cycle ergometer (in a sitting position) with resistance equal to 7.5% of body weight and, apart from modified (shortened) performance time, it was carried out in accordance with the Wingate test methodology (1). The subjects commenced their trials in a still position. The trials were preceded by a 2-minute warm-up performed on

a cycle ergometer with no resistance (13). After a 2-minute rest period each participant began the test.

The 5 *alternate leg bounds test* was performed on a runway and long jump sand pit. Each subject started her jump from a line placed 8–9 m from the pit. After 5 bounds the subjects landed with both feet in the pit. Each subject was expected to jump as far as possible (9).

Countermovement jumps and UCMJs were performed on a Kistler piezoelectric force platform (Kistler 9281CA, Winterthur, Switzerland) with sampling frequency of 500 Hz. Signals from the platform were amplified and recorded on a PC computer using a 16-bit A/D board and Bio Ware 3.24 software. Peak power in the concentric phase was calculated as the largest vertical ground reaction force F_z and vertical velocity v_z product (Figure 1). The vertical velocity was obtained by numerical integration of the vertical acceleration extracted from the vertical ground reaction force (5). While performing CMJ and UCMJ tests, the subjects kept their hands on the hips. The knee joint was flexed at approximately

TABLE 2. Plyometric exercise program.*

Week	Exercise program for ULE group and BLE group (set $\times$ repetition) [†]
1–2	Side-to-side ankle hop over a slat 4 $\times$ 8 Standing rope jumps 8 $\times$ 15 Zig-zag jumps 4 $\times$ 8 Horizontal leg bounds 4 $\times$ 6 Horizontal backward skip 2 $\times$ 6 Hurdle jumps (ULE–0.20 m; BLE–0.35 m) 4 $\times$ 6
3–4	Side-to-side ankle hop over a slat 4 $\times$ 8 Standing rope jumps 8 $\times$ 15 Horizontal leg bounds 4 $\times$ 8 Horizontal backward skip 2 $\times$ 8 Hurdle jumps (ULE–0.20 m; BLE–0.35 m) 6 $\times$ 4 Step jumps [‡] (ULE–every step, BLE–every 2 steps) 4 $\times$ 8 Jump onto a box (ULE–0.15 m; BLE–0.25 m) and jump-off backwards 2 $\times$ 8
5–6	Standing rope jumps 4 $\times$ 15 Rope jumps forwards 4 $\times$ 15 Horizontal leg bounds 4 $\times$ 10 Lateral hurdle jumps (ULE–0.20 m; BLE–0.35 m) 4 $\times$ 4 Hurdle jumps (ULE–0.20 m; BLE–0.35 m) 6 $\times$ 6 Step jumps (ULE–every step, BLE–every 2 steps) 6 $\times$ 8 Jump onto a box (ULE–0.20 m; BLE–0.35 m) and jump-off backwards 6 $\times$ 8
7–8	Drop jumps without take-off (ULE–0.30 m; BLE–0.50 m) 2 $\times$ 4 Standing rope jumps 4 $\times$ 15 Rope jumps forwards 4 $\times$ 15 Standing vertical jumps 4 $\times$ 4 Horizontal leg bounds uphill 4 $\times$ 10 Lateral hurdle jumps (ULE–0.25 m; BLE–0.45 m) 4 $\times$ 4 Hurdle jumps (ULE–0.25 m; BLE–0.45 m) 6 $\times$ 6 Step jumps (ULE–every 2 steps, BLE–every 3 steps) 8 $\times$ 8 Drop jumps without take-off (ULE–0.35 m; BLE–0.60 m) 4 $\times$ 4
9–10	Standing rope jumps 4 $\times$ 15 Rope jumps backwards 4 $\times$ 15 Horizontal leg bounds downhill 4 $\times$ 8 Standing vertical jumps 2 $\times$ 4 Standing vertical jumps with the overhead goal 2 $\times$ 4 Hurdle jumps (ULE–0.30 m; BLE–0.50 m) 4 $\times$ 6 Step jumps (ULE–every 2 steps, BLE–every 3 steps) 8 $\times$ 8 Drop jumps with a vertical jump (ULE–0.15 m; BLE–0.30 m) 4 $\times$ 4
11–12	Standing rope jumps 6 $\times$ 15 Horizontal leg bounds 4 $\times$ 10 Standing vertical jumps with the overhead goal 4 $\times$ 4 Vertical jumps with a 2-stride approach with the overhead goal 12 $\times$ 1 Hurdle jumps (ULE–0.30 m; BLE–0.50 m) 6 $\times$ 6 Drop jumps with a vertical jump with the overhead goal (ULE–0.20 m; BLE–0.35 m) 4 $\times$ 4

*ULE = unilateral plyometric group; BLE = bilateral plyometric group.

[†]2 $\times$ 10 jumps performed by the BLE group were equivalent to performing 1 $\times$ 10 right-leg jumps + 1 $\times$ 10 left-leg jumps by the ULE group.

[‡]Step height was 0.25 m.

calculated by taking the average of best trials for a given extremity.

Before CMJ, UCMJ, and 5 alternate leg bounds, the subjects were instructed to perform a low-intensity 8-minute jog as a warm-up exercise followed by 5-minute dynamic stretching exercises, 3 $\times$ 20-m skipping, and unilateral (1 $\times$ 8) and bilateral (1 $\times$ 8) rebounds.

Training Procedure

Before the main program, the subjects participated in 6 remedial training sessions to learn a proper technique of plyometric exercises. The experimental training program lasted 12 weeks with 1 week of no training during the sixth week during which time the midtraining testing was performed. The training sessions were held on Mondays and Thursdays. Apart from step jumps, all the other jumps were performed on a natural grass surface. During 40- to 45-minute sessions, the participants performed low-intensity runs (5 minutes), dynamic stretching exercises (5 minutes)—swings, rotations, bends, and plyometric exercises. The ULE group performed unilateral jumps only, whereas the BLE group performed bilateral jumps (Table 2). Both training groups performed the same number of contacts. To equalize the intensity of unilateral and bilateral exercises, jump and obstacle heights in the ULE group were decreased to 55–60% of the height used in the BLE group, and in the case of step jumps, it was lowered to 50%. This procedure was based on study

90°, and the free leg swing and trunk bending movements were not allowed during UCMJ execution. Each subject was expected to jump as high as possible. The UCMJ height was

results (24,26,27) and our observations made during the pilot studies, when we compared impulses of the reaction force during the concentric phase of 1-leg and 2-leg jumps.

TABLE 3. Effects of plyometric training on peak power and jumping ability (mean $\pm$ SD) in alternate leg tests.*

	ULE	BLE	CON
Ergometric test (W)			
Pretraining	514 $\pm$ 61	527 $\pm$ 52	521 $\pm$ 64
Midtraining	575 $\pm$ 59 ^{†‡}	561 $\pm$ 57 [‡]	527 $\pm$ 67
Posttraining	599 $\pm$ 55 ^{†‡}	612 $\pm$ 49 ^{†‡§}	534 $\pm$ 73
Detraining	547 $\pm$ 61 ^{‡§}	561 $\pm$ 45 [‡]	526 $\pm$ 58
Five alternate leg bounds (m)			
Pretraining	10.55 $\pm$ 0.40	10.42 $\pm$ 0.43	10.39 $\pm$ 0.42
Midtraining	10.89 $\pm$ 0.42 ^{†‡}	10.58 $\pm$ 0.37	10.43 $\pm$ 0.33
Posttraining	10.96 $\pm$ 0.39 ^{†‡}	10.76 $\pm$ 0.34 ^{†‡}	10.35 $\pm$ 0.28
Detraining	10.73 $\pm$ 0.31 ^{†‡§}	10.64 $\pm$ 0.29 [‡]	10.33 $\pm$ 0.30

*ULE = unilateral plyometric group; BLE = bilateral plyometric group; CON = control group.

[†]Significantly different ($p < 0.05$) from control group measurements.

[‡]Significantly different ($p < 0.05$) from pretraining measurements.

[§]Significantly different ($p < 0.05$) from previous measurements.

Exercise intensity was increasing in a linear manner until the end of the program. After each session, the subjects were hanging down the wall bars 3×5 seconds and performed abdomen and back exercises 2×10 repetitions. All the classes were supervised by the same instructor to ensure consistency, and the participants were asked not to do any other physical exercises during the study.

Statistical Analyses

The data were initially tested for normality and homogeneity of variance assumptions. Because the assumptions were not violated, 1-way analysis of variance (ANOVA) was conducted

1.2, respectively. The level of statistical significance was set at $p \leq 0.05$. Statistica v. 5.1 was the program, which was used for calculations.

RESULTS

Alternate Leg Tests

The alternate leg test results are given in Table 3. The applied training program resulted in significant changes in Wingate peak power and jumping ability in alternate tests (group $\times$ time interaction was significant: $F_{6,138} = 11.03$, $p < 0.001$ and $F_{6,138} = 7.57$, $p < 0.001$, respectively). Peak power increased significantly ($p < 0.05$) by 11% in the ULE group (ES = 1.0) and 6% in the BLE group (ES = 0.6) from pretraining to midtraining. The ULE group did not demonstrate any significant improvement in peak power in posttraining, whereas in the BLE group, peak power significantly increased ($p < 0.05$) by another 8% (ES = 0.9). Peak power in the ULE group decreased substantially ($p < 0.05$; ES = 0.9) in detraining, whereas in the BLE group, no significant change was observed. Peak power values of both groups were still significantly higher ($p < 0.05$) than those measured pretraining.

Both training groups achieved better results in the 5 alternate leg bounds test. This improvement ($p < 0.05$) occurred after 6 weeks

TABLE 4. Effects of plyometric training on peak power and jumping ability (mean $\pm$ SD) in bilateral leg tests.*

	ULE	BLE	CON
CMJ (W)			
Pretraining	2,059 $\pm$ 139	2,078 $\pm$ 123	2,021 $\pm$ 110
Midtraining	2,181 $\pm$ 149 ^{†‡}	2,149 $\pm$ 128 [‡]	2,040 $\pm$ 116
Posttraining	2,207 $\pm$ 128 ^{†‡}	2,223 $\pm$ 116 ^{†‡§}	2,027 $\pm$ 121
Detraining	2,134 $\pm$ 126 ^{‡§}	2,176 $\pm$ 107 [‡]	2,051 $\pm$ 109
CMJ (m)			
Pretraining	0.23 $\pm$ 0.04	0.22 $\pm$ 0.03	0.21 $\pm$ 0.03
Midtraining	0.27 $\pm$ 0.04 ^{†‡}	0.25 $\pm$ 0.04 ^{†‡}	0.22 $\pm$ 0.04
Posttraining	0.27 $\pm$ 0.03 ^{†‡}	0.27 $\pm$ 0.03 ^{†‡}	0.22 $\pm$ 0.03
Detraining	0.26 $\pm$ 0.04 ^{†‡}	0.25 $\pm$ 0.03 ^{†‡}	0.21 $\pm$ 0.03

*ULE = unilateral plyometric group; BLE = bilateral plyometric group; CON = control group; CMJ = countermovement jump.

[†]Significantly different ($p < 0.05$) from control group measurements.

[‡]Significantly different ($p < 0.05$) from pretraining measurements.

[§]Significantly different ($p < 0.05$) from previous measurements.

TABLE 5. Effects of plyometric training on peak power and jumping ability (mean $\pm$ SD) in unilateral leg tests.*

	ULE	BLE	CON
UCMJ (W)			
Pretraining	1,457 $\pm$ 134	1,440 $\pm$ 126	1,431 $\pm$ 121
Midtraining	1,548 $\pm$ 143 ^{†‡}	1,490 $\pm$ 114	1,449 $\pm$ 127
Posttraining	1,533 $\pm$ 131 [‡]	1,519 $\pm$ 137 [‡]	1,467 $\pm$ 109
Detraining	1,497 $\pm$ 138	1,479 $\pm$ 123	1,453 $\pm$ 115
UCMJ (m)			
Pretraining	0.13 $\pm$ 0.02	0.14 $\pm$ 0.03	0.12 $\pm$ 0.03
Midtraining	0.16 $\pm$ 0.03 ^{†‡}	0.16 $\pm$ 0.03	0.13 $\pm$ 0.02
Posttraining	0.18 $\pm$ 0.03 ^{†‡§}	0.17 $\pm$ 0.02 ^{†‡}	0.13 $\pm$ 0.03
Detraining	0.17 $\pm$ 0.03 ^{†‡}	0.16 $\pm$ 0.02 [†]	0.13 $\pm$ 0.04

*ULE = unilateral plyometric group; BLE = bilateral plyometric group; CON = control group; UCMJ = unilateral countermovement jump.

[†]Significantly different ($p < 0.05$) from control group measurements.

[‡]Significantly different ($p < 0.05$) from pretraining measurements.

[§]Significantly different ($p < 0.05$) from previous measurements.

in the ULE group (4%; ES = 0.9), whereas in the BLE group after 12 weeks (3%; ES = 0.8). Posttraining gains in both groups were significantly higher ($p < 0.05$) than in the CON group (ES = 1.8 and ES = 1.2, respectively). After the detraining condition, the BLE group maintained their 5 alternate leg bounds performance, whereas the ULE group significantly decreased (2%; ES = 0.6) posttraining gains. However, both groups demonstrated significantly higher ($p < 0.05$) jumping ability than the CON group did (ES = 1.4 and ES = 1.1).

Bilateral Leg Tests

The bilateral tests results are given in Table 4. They indicate that the training program improved peak power and jumping ability in both groups (group $\times$ time interaction was significant: $F(6,138) = 14.02$, $p < 0.001$ and $F(6,138) = 6.76$, $p < 0.001$). They considerably increased their peak power (6%, ES = 0.9 for ULE and 3%, ES = 0.6 for BLE) from pretraining to midtraining. After 12 weeks of training, peak power increased by another 4% (ES = 0.6) in the BLE group and remained unchanged in the ULE group. Detraining decreased ($p < 0.05$) peak power in the ULE group by 3% (ES = 0.6), whereas this parameter did not alter in the BLE group. The last 3 measurements show that peak power improvement in both experimental groups was significantly higher ($p < 0.05$) than in the CON group (ES = 1.4, 1.7, 0.9 for ULE and 1.1, 1.8, 1.4 for BLE).

The experimental groups improved significantly their jumping ability from pretraining to midtraining (14%, ES = 0.9 for ULE; 11%, ES = 1.0 for BLE). All measures of this parameter were still above pretraining values during the succeeding measurements. These results were significantly higher ($p < 0.05$) compared to those of the CON group (ES = 1.7, 1.3, 1.8 for ULE and ES = 1.0, 1.3, 1.7 for BLE).

Unilateral Leg Tests

The unilateral tests results are given in Table 5. They show considerable differences in peak power and jumping ability among the groups (group $\times$ time interaction was significant: $F(6,138) = 3.86$, $p < 0.01$ and $F(6,138) = 7.00$, $p < 0.001$). Only the ULE group considerably improved in peak power from pretraining to midtraining (4%; ES = 0.4). After the completion of the whole training program, peak power remained at midtraining values in the ULE, and significantly increased in the BLE group (4%; ES = 0.5) in comparison with pretraining values. These values did not differ significantly ($p > 0.05$) from those observed in the

CON group. All the 3 groups demonstrated the same level of maximum power during the last measurements.

The ULE group improved significantly ($p < 0.05$) in jumping height (19%; ES = 1.1) from pretraining to midtraining. Both experimental groups significantly improved their jumping scores after 6 weeks ($p < 0.05$), but the ULE group also demonstrated an increase after 12 weeks. Four weeks after the completion of the program, the jumping ability in both groups did not differ significantly from the previous measurements and was still considerably higher ($p < 0.05$) than in the CON group (ES = 1.8 for ULE and ES = 1.5 for BLE).

DISCUSSION

The major findings of this study are that changes in peak power and jump performance during 12 weeks of training and 4 weeks of detraining were determined by the type of plyometric training performed in untrained women. During the first 6 weeks of unilateral plyometric training, there was an increase in all power and jumping ability tests, which was manifested unilaterally, bilaterally, and alternately, whereas bilateral plyometric training produced significant improvement in bilateral tests only. However, during the following 6 weeks, the peak power and jumping ability did not change in the ULE group, whereas they continued to increase in the BLE group. The large ES noted between plyometric groups and the control group during posttraining measures might be because of training intervention and poor starting performance of the latter group. Four weeks of detraining induced a decreasing tendency for power and jumping ability in alternate leg tests in the ULE group and maintenance of power and jumping ability in the BLE group. This observation has some potentially interesting implications

for the implementation of plyometric training in terms of the timing of stimuli throughout a competitive season.

Based on previous studies (4,23), it was expected that power production and a jumping performance during the first 6 weeks would increase more after unilateral jumps than bilateral ones. In examining possible mechanisms for this improvement, it is logical to suggest that greater gains in the 5 alternate leg bounds and UCMJ tests in the ULE group may be related to more effective recruitment of the muscles. Van Soest et al. (26) reported that electromyographic activity of vastus medialis and gastrocnemius during the concentric phase of the CMJ was 10–20% lower when compared to the UCMJ. A higher activity of stabilizing muscles also occurs in unilateral jumps because of a center of gravity which is not in line with the base of support. The increased activation of stabilization musculature has been suggested to improve the orientation of the head, arms, and trunk with respect to the lower extremities during jumping and landing tasks. Furthermore, previous researchers have reported this to attenuate the ground reaction force absorption during landing (14) and provide better results in unilateral and alternate tests (19).

In the second part of the program, no significant rise in values of researched parameters was observed in the ULE group. A linear trend of the increase of power and jumping ability was observed in the BLE group during the 12 weeks' intervention. As a result, when the program was completed, a total gain of power and jump performance was identical in both experimental groups despite the more rapid gain in the ULE group in the first 6 weeks. In alternate and bilateral tests, the BLE group reached even a higher (though statistically insignificant) level than the ULE group did. The results achieved after the training support the observations described by Stannard (23) and McCurdy et al. (18). However, the difference in time course of performance changes in both plyometric groups may indicate a different underlying mechanism responsible for power improvement and jumping ability between the 2 training modalities employed in this investigation. Contact time during unilateral jumps is longer when compared to bilateral jumps, which leads to some leg extensors shortening at higher velocities in unilateral jumps. This suggests a lower level of force production because of the force–velocity relationship (2). Because of this relationship, we suggest that our unilateral training program provided a greater strength stimulus, whereas the bilateral training program had a greater impact on movement velocity. Furthermore, it is likely that these were significant factors which influenced the difference in time course of performance changes between the ULE and BLE groups. This observation can be partly supported by the findings of Markovic et al. (16), which did not show any improvement in legs strength after a 10-week bilateral plyometric training program, whereas unilateral sprint exercises did. It is also noteworthy that there was a significant improvement in all specific tasks (i.e., UCMJ for the ULE and CMJ for the BLE group) during the first half of the training program, corroborating the principle of specificity (21).

The effectiveness of plyometric training can also be estimated on the basis of maintaining a high performance with associated reduction or cessation of training stimulus (detraining). In this context, the training program completed by the BLE group was more effective, as the level of power and jumping ability remained unchanged when compared to measurements taken immediately after the training. This supports the results of Ebben et al. (6) that training gains can be maintained at least for 10 days, and of Marques and González-Badillo (17), who demonstrated a training retention of 2–6 weeks, depending on the training status. Herrero et al. (10) claim that they may even get higher during 2 weeks after the training completion which certainly seems probable given the duration of tapering cycles showing improvements in other performance parameters along similar timelines. Finally, it also seems that the significant loss of power and jumping ability in the alternate test and the decrease of power in the bilateral test in the ULE group during detraining originate from the lack of physiological adaptation to long-term unilateral activity.

PRACTICAL APPLICATIONS

The results from this study suggest that both unilateral and bilateral forms of plyometric exercise are valuable and should be considered by the strength and conditioning professional. The issue between when to use one or the other appears to be one of timing and specific goals, which may cause the practitioner to emphasize one form of training over the other. For example, it appears that performing unilateral plyometric exercises is more effective at increasing measures of power output during a short-term training cycle compared to bilateral plyometric exercises. Therefore, a training program involving a greater emphasis on unilateral exercises is especially recommended, if athletes need to increase power output several times during a macrocycle, particularly before peaking for major competitions throughout a season. In addition, athletes who have little time for whatever reason to prepare for a specific event may improve performance more rapidly through the use of unilateral plyometric exercise.

However, bilateral plyometric regime would seem to guarantee maintenance of high power performance after reduction or cessation of explosive exercises, at least for 4 weeks, whereas unilateral exercise did not. In sports where the in season schedule may preclude frequent training or, during times of active recovery, bilateral exercises may be more effective in preventing detraining from a power standpoint. In these situations, bilateral forms of plyometric training should be considered when designing a training and conditioning program with the intent of improving or maintaining power in sports with a longer competition period. In addition, bilateral exercises may be considered during the off-season period where more time exists to make gains exist.

Because plyometrics is a common training modality, the conclusions drawn from this study are directly applicable for

untrained subjects and young athletes not involved in strength training. More study is recommended to discern the optimal manner to incorporate these methods for those who are participating in more traditional forms of resistance training at the same time.

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